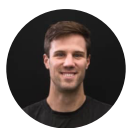


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June 17, 2024

The science of real-time bar speed feedback

The Metric VBT app now features the ability to enable real-time feedback for even more velocity based training applications in your program.

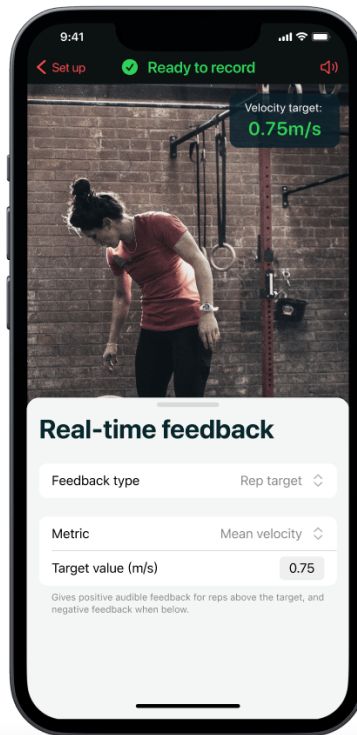


Jacob Tober
Co-founder

The Metric VBT app now features a real-time feedback feature for an expanded range of velocity based training uses in your training and programming. You can read more about the specifics of this new feature [here](#), but for now, let's dive into the scientific support of real-time feedback, and how you can apply this in your training and



coaching.



Feedback is scientifically proven to enhance strength and power

At its core, velocity based training is built around the benefits of real-time feedback. More than just a simple motivational tool, feedback encourages greater intent and effort in every rep, motivating you to give everything you can in a set.

Real-time feedback is a proven and effective way to enhance strength and power training adaptations. When used well, velocity feedback improves acute performance during a training session – increasing vertical leap by 4% in a single session. Feedback also improves athletic transfer, giving on-field performance a boost thanks to higher quality training. In some studies, this boost is as much as 2x the improvements

seen in sprinting speed and jump height compared to non-feedback training.

Results from
a 2019 study.
Velocity
based
training
feedback
improves
performance
in training.

There are three types of velocity feedback, each with specific applications and benefits — Announce each rep, velocity target/zones, and the ability to set a velocity loss threshold. You can also choose between mean velocity, peak velocity, range of motion, and eccentric tempo for your feedback focus.

Let's dive into how you can apply each of these feedback styles in your training.

Announce each rep

With this setting enabled, Metric will audibly announce the chosen data point at the top of each rep, motivating you to adjust your next rep in response. We've fine-tuned the feedback to be focused and helpful, rounding metrics to fewer decimals and skipping the "zero" for reps below 1.0m/s for a faster and more focused feedback experience.

Example applications of rep announcement:

- **Calibrating tempo.** Getting the correct eccentric tempo can be tricky, by using rep announcements you can adjust after every rep to ensure your athletes are hitting the prescribed eccentric tempos.
- **Dialling in ROM.** Getting a consistent range on exercises like RDLs and squats can be tricky without the feedback of touching the bar to the chest as with a bench press or ground as with conventional deadlifts. Use the ROM feedback to refine your technique and help you get a sense for when your range hits or misses the desired target.
- **Chasing speed (or height) on jumps.** Enable peak velocity feedback during your trapbar jumps to see if you can hit higher and higher bar speeds – a great way to create competition in the gym between athletes or simply fight the fatigue effects of a hard training session. Feedback like this is a great way to get an immediate boost in training performance during team sessions. Watch these numbers go up and up once teammates start pushing each other to do better!

Velocity target

Setting a velocity target, or zone, provides lifters with a clear performance goal and specific focus for that set. This use of velocity tracking data is one of the oldest and simplest applications for feedback, allowing athletes to simply work hard and chase the positive alerts when reps hit the optimal velocity zone or increasing the intent and adding motivation in order to avoid a negative alert if velocity falls

short.

This simple type of feedback is perfect when training with velocity zones, or if you are trying to encourage higher intent and explosive reps in dynamic effort training.

Example applications of velocity targets:

- **Target desired velocity zones** – Encourage athletes to self-select a load that allows them to keep all reps above a given value. Targets of 1.0m/s or 0.75m/s are commonly used to target dynamic effort power training (commonly called speed-strength or strength-speed). An even better option is to link the velocity target with an athletes 6-week average velocity for that weight (a number calculated automatically for every set in the Metric VBT app)
- **Last rep velocity cutoff** – Use a specific velocity target to have athletes complete as many reps as possible on a weight until they fall below the target, ensuring sets end at the goldilocks exertion levels you are chasing. As a guide, the lower the last rep cut-off, the harder the set is going to be!

Velocity loss thresholds

Setting a desired velocity loss percentage, ranging from 5% to 45% is a great way to autoregulate the number of reps in a set and the proximity to failure of that set.

This feedback only activates as you approach the specified threshold,

allowing you to push the set hard and having Metric chime in to let you know when the velocity loss threshold is approaching, ensuring the ideal proximity to failure is hit with every set.

Many studies have shown that training with a 20-25% velocity loss (equivalent to an ~8 RPE) in strength training is superior to develop strength compared to sets taken to 35-40% velocity loss (RPE 9.5 / 10).

Despite completing less total training volume, training with a 20-25% threshold led to better gains in strength and power – due to lower accumulating fatigue and faster recovery between sessions.

A 2016 study
on velocity
loss.

Training
with only
20% velocity
loss was
superior to
40% velocity
loss for
training
adaptations,
despite
performing
less training
volume.

Example applications of velocity loss thresholds:

- **Autoregulated open-rep count sets** – Set a velocity loss target and continue performing reps until the velocity loss threshold is passed. The higher the velocity loss target the tougher the set and

the more fatiguing it will be!

- **Controlled proximity to failure** – A great way to push athletes and help them really find their physical limits is to set a high velocity loss threshold (say 30–35%) and have them rep out a set, only stopping the set when they do finally reach the threshold. 35% is not quite complete failure, and some lifters might surprise themselves unlocking more reps when challenged to compete like this.
- **Keep power output high** – Setting a tighter threshold of ~10% might be useful during power training or for a tapering phase. A lifter will aim to complete a fixed number of reps on a power exercise (say 3–5) but using this tighter velocity loss threshold they will end a set early if the threshold is passed. Using this approach the coach can program a “total” number of reps on a power exercise, using as many sets as required while keeping velocity loss to a minimum. A great way to maintain quality of reps and a high power output during the session.

When to Opt Out of Velocity Feedback

While real-time feedback is a potent training tool, there are times when you might want to leave it deactivated and just focus on finishing the set uninterrupted.

Avoid using real-time feedback during heavy, low-rep strength sets of 1, 2, or 3 reps. For these big sets its best to simply lock in, complete the task and analyse the data afterwards. Similarly, in early learning phases of new movements, audible feedback may be more distracting than beneficial so be selective of when to enable this for novice lifters.

Lastly, if you are the kind of lifter who is easily distracted in general, real-time feedback might be a net-negative in your training, it might be smart to switch it off and rely on post-set data review instead.

[Give real-time feedback a try in the Metric VBT app \(iOS\)→](#)

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